Instructor: Justin Baker, `justin@math.ucla.edu`

Lectures Time & Room: MWF 2pm-2:50pm, Geology Building 6704

Course website: `https://bruinlearn.ucla.edu/courses/215961`

Office Hours: Mondays 10-11am, MS 7324 (or by appointment)

Textbooks: Burden, R.L. and Faires, J.D. *Numerical Analysis* Cengage Learning, 2010

Prerequisite: Math 32B, 33B and 115A and some experience with scientific computing.

Course Objectives: This course will introduce the fundamental algorithms and analysis techniques of numerical methods that form the backbone of scientific computing. While the course emphasizes theoretical principles such as stability, convergence, and error analysis, these will be closely tied to practical implementation.

- ▹ Develop an understanding of key numerical algorithms for solving equations, interpolation, numerical integration, and differential equations.
- ▹ Apply these algorithms to real-world scientific and engineering problems, interpreting accuracy, stability, and computational efficiency.
- ▹ Communicate insights and explore extensions of classical numerical methods to new applications.

Grading policy: The grade for the class will be based on seven assignments, two midterm exams, and one final exam. The scores for each of these grade components will be weighted according to the following weighting scheme:

Final Exam 20%, Midterm Exams 20%, Assignments 60%.

- Any re-grade requests must be submitted within one week of receiving your score.
- Note that you must take the final exam in order to pass this class.
- The letter grade assignments will be given after the final exam; however,

90 or higher	guarantees an A- or better letter grade;
80 or higher	guarantees a B- or better letter grade;
70 or higher	guarantees a C- or better letter grade;
60 or higher	guarantees a D- or better letter grade.

- You are expected to follow the UCLA Student Conduct Code, which can be found here: `https://www.deanofstudents.ucla.edu/Individual-Student-Code`. If you are not familiar with it, take a few minutes to read about your responsibilities. Violations will be referred to the Dean of Students.

Schedule of Lectures (tentative)			
Week	Day	Book	Topic
1	F	1.1	General course overview and machine numbers
2	M	1.2	Errors
	W	1.3	Algorithms and convergence (A1 Assigned)
	F	2.1	The bisection method
3	M	2.2	Fixed-point iteration
	W	2.3	Newton's method, (A1 Due)
	F	2.3	Secant method, and method of False Position
4	M	2.4	Convergence order. Multiple roots
	W	2.5	Accelerating convergence (A2 Due)
	F	2.6	Zeros of polynomials. Horner's method
5	M		**Midterm 1: Monday, October 20 in Class**
	W	2.6,3.1	Deflation and Lagrange polynomials (A3 Assigned)
	F	3.1,3.2	Lagrange polynomials and Neville's method
6	M	3.3	Divided differences, interpolation nodes and finite difference
	W	343	Hermite Interpolation(A3 Due)
	F	3.5	Cubic spline interpolation
7	M	4.1	Forward/backward difference
	W	4.1	Finite-difference formulas (A4 Due)
	F	4.2,4.3	Richardson's extrapolation. Interpolation based numerical integration
8	M		Veterans Day
	W	4.3,4.4	Newton-Cotes formulas. Composite integration formulas (A5 Due)
	F	4.5	Romberg integration
9	M		**Midterm 2: Monday, November 17 in Class**
	W	4.7	Gaussian quadrature(A6 Assigned)
	F	6.1	Solving linear systems
10	M	6.2	Pivoting
	W	6.6	Special types of matrices (A6 Due)
	F		*Thanksgiving Holiday*
11	M	7.1,7.3	Review of matrix algebra. Jacobi's method
	W	7.3	Gauss-Seidel method (A7 Due)
	F		Review
	Finals		**Tuesday, December 9, 8:00am-11:00am, Room TBD**

Assignments: Your solutions to the exercises should clearly demonstrate your understanding of your solution. For all coursework other than exams, you are allowed and indeed strongly encouraged to work with your classmates. Work that you submit should be your own work; don't simply copy someone else's work line for line. Forming study groups to collaborate on the assignments and studying for exams is an excellent way to learn, and is encouraged. You are to submit your assignments using GradeScope. Please include in your assignments' submission who you worked with and any external resources used.

Accomodations: Students needing academic accommodations should contact the Center for Accessible Education (CAE) at (310)825-1501 or in person at Murphy Hall A255. In order to ensure accommodations, students need to contact the CAE within the first two weeks of term. Please let me know if you have any other accommodations as soon as possible and do not hesitate to reach out.

Classroom Expectations: All members of this class are expected to contribute to a respectful, welcoming, and inclusive environment for every other member of the class. You are expected to act respectfully toward others, whatever our age, background, belief, race, ethnicity, gender, gender identity, gender expression, sexual orientation, national origin, religious affiliation, ability, and other visible and non-visible differences.

Classroom Agreements: These agreements are also applicable to discussion sessions and group work:

1. We're all learning.
2. View mistakes and discomfort as opportunities for growth.
3. The goal is to be intelligible to every person in the class. Cite your sources, explain cultural references, and define your terms to bring everyone along with you.
4. Respect and honor identities and preferences (e.g. learn and use names and pronouns).
5. Give space for others to share (give space, take space).
6. Challenge the idea, not the person.
7. Recognize how your own social positionality (e.g., race, class, gender, sexual orientation ability, ...) informs your perspectives.
8. Remember everyone comes in with different life and academic experiences than yours. No one is better than another.
9. Understand and appreciate different approaches to problem solving.

Academic Integrity: From the office of the Dean of Students:

> *With its status as a world-class research institution, it is critical that the University uphold the highest standards of integrity both inside and outside the classroom. As a student and member of the UCLA community, you are expected to demonstrate integrity in all of your academic endeavors. Accordingly, when accusations of academic dishonesty occur, The Office of the Dean of Students is charged with investigating and adjudicating suspected violations. Academic dishonesty, includes, but is not limited to, cheating, fabrication, plagiarism, multiple submissions or facilitating academic misconduct.*

Students are expected to be aware of the University policy on academic integrity in the UCLA Student Conduct Code[1]. In particular, please note the sections on (1) cheating, (2) plagiarism, and (3) unauthorized study aids.

[1] `http://www.deanofstudents.ucla.edu/Portals/16/Documents/UCLACodeOfConduct_Rev030416.pdf`

Mental Health Support: Mental health is health. If you are going through a hard time please know there is help available. Counseling and Psychological Services (CAPS) has trained professionals who can help. Contact (310) 825-0768 or go to `https://counseling.ucla.edu/services/are-you-in-a-crisis`. If you are concerned about a friend or peer, discuss your concerns with a professional who can discreetly connect the distressed individual with the proper resources (`https://counseling.ucla.edu/concerned-about-a-bruin/concerned-about-a-bruin`). Reach out and ask for help if you or someone you know is having a difficult time.

Other Resources and Support: Please check out `https://docs.google.com/document/d/1K086I03xtvgumRBio1sA8pVfpeb-w-r4TPMvbcsJ0tw/edit`.

Notice about sexual harassment, discrimination and assault: Title IX prohibits gender discrimination, including sexual harassment, domestic and dating violence, sexual assault and stalking. Students who have experienced sexual harassment or sexual violence can receive **confidential** support and advocacy from a CARE advocate:

> The CARE Advocacy Office for Sexual and Gender-Based Violence
> 1st Floor, Wooden Center West
> `CAREadvocate@caps.ucla.edu`
> (310) 206-2465

You can also report sexual violence or sexual harassment directly to the University's Title IX Coordinator:

> Kathleen Salvaty
> 2241 Murphy Hall
> `titleix@conet.ucla.edu`
> (310) 206-3417